Sandeep Kumar

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PROFESSIONAL SUMMARY

PhD researcher at IIT Bombay specializing in Computer Vision, Medical Image Analysis, and Vision-Language Models. Experienced in developing end-to-end deep learning pipelines for medical imaging (CT/MRI), 2D/3D CNN architectures, and segmentation models including U-Net and nnU-Net. Strong background in machine learning research, teaching at top institutions, and mentoring graduate-level students. Proficient in Python, PyTorch, TensorFlow, and scikit-learn.

RESEARCH INTERESTS

Computer Vision, Medical Image Analysis, Vision-Language Models (VLMs), Machine Learning, Deep Learning, Multimodal Learning

EDUCATION

Indian Institute of Technology Bombay, Mumbai, India — 2024 – Present
PhD, Center of Machine Intelligence and Data Science
CGPA: 8.46
Relevant Coursework: Introduction to Machine Learning, Advanced Topics in Machine Learning, Deep Learning for NLP, Deep Learning for Computer Vision, Engineering Statistics, Optimization, Linear Systems, Mathematical Structures for Control.

Central University of Rajasthan, Rajasthan, India — 2018 – 2023
Integrated Master's in Computer Science
CGPA: 8.40
Relevant Coursework: Operating Systems, Analysis of Algorithms, Artificial Intelligence, Machine Learning, Data Mining, Natural Language Processing, DBMS, Compiler Design, Probability and Statistics, Network Security.

RESEARCH EXPERIENCE

Research Intern — May 2025 – Feb 2026
Transformer Neural Scan Quest Pvt. Ltd. (TNSQAI), Remote

- Developed multi-class classification and segmentation pipelines for brain abnormality detection from CT scans, supporting computer-aided medical diagnosis.
- Curated medical imaging datasets in DICOM and NIfTI formats; built preprocessing pipelines including brain and subdural windowing, skull stripping, intensity normalization, and PHI anonymization.
- Built and evaluated 2D and 3D CNN-based pipelines along with U-Net and nnU-Net variants for medical image analysis tasks.
- Addressed class imbalance using Dice loss, Focal loss, data augmentation strategies, and cross-validation for robust model development.
- Tracked experiments and evaluated model performance using Dice score, IoU, sensitivity, specificity, and AUC; documented findings for research reporting and deployment handoff.

TEACHING EXPERIENCE

Teaching Assistant — Jan 2025 – Present
Indian Institute of Technology Bombay, Mumbai, India

- Teaching Assistant for EE 782 (Advanced Topics in Machine Learning), DS 303 (Introduction to Machine

Learning), and EE 769 (Introduction to Machine Learning).
- Conducted tutorials, resolved student doubts, assisted with assignments, and supported evaluation for undergraduate and postgraduate students.
- Helped students build conceptual and practical understanding of machine learning foundations, model development, and experimental analysis.

Teaching Assistant - Deep Learning and Generative AI Jan 2026 – Present
Great Learning and IIT Bombay, India

- Provided teaching support for a Deep Learning and Generative AI program including learner guidance, technical mentoring, and concept clarification.
- Assisted participants with neural networks, deep learning workflows, model training, and emerging generative AI techniques.

Assistant Professor, Computer Science Sep 2023 – Jan 2024
Central University of Rajasthan, Ajmer, India

- Delivered lectures, mentored students, and supported academic instruction in computer science courses.
- Contributed to curriculum refinement and departmental academic activities.
- Guided student projects and provided mentorship related to higher studies and career development.

Research and Project Mentor 2025 – Present
Student Mentorship and Academic Guidance, India

- Mentored 12 B.Tech. and M.Tech. students in academic and research projects.
- Guided students through problem formulation, literature review, implementation, experimentation, result analysis, and technical report preparation.

PUBLICATIONS

- Bugalia, S.; Kumar, S.; Tripathi, J.P.; Martcheva, M. Dynamic Interplay of Allee Effect and Harvesting in a Diseased Amensalism Model: Unraveling Ecological Complexity. Mathematics and Computers in Simulation, 2026.

SELECTED PROJECTS

Medical Image Analysis for Brain Abnormality Detection 2025

- Developed deep learning pipelines for CT-based brain abnormality classification and segmentation using clinically relevant preprocessing and evaluation protocols.
- Worked with 2D and 3D architectures and segmentation models for improved abnormality localization and diagnostic support.

Vision-Language Modeling for Medical and Multimodal Understanding 2025

- Explored multimodal learning workflows integrating visual features and language representations for image-text understanding and semantic reasoning tasks.
- Investigated the use of Vision-Language Models (VLMs) for improved contextual interpretation in AI systems.

Industrial Visual Anomaly Detection for Automated Inspection 2025

- Designed an anomaly detection pipeline for industrial visual inspection using deep feature learning and image-based defect localization.

- Worked on identifying abnormal patterns in manufacturing data to support automated quality control and reduce manual screening effort.

Human Activity Recognition using Smartphone Sensor Data 2022

- Developed a machine learning model to predict user activities using smartphone sensor data such as gyroscope and accelerometer signals.

Winery Prediction and Model Serving API 2023

- Built a predictive model for wine variety classification using customer review data and implemented an API for model serving.

Bitcoin Trading Bot using Time-Series Learning 2023

- Designed and implemented a Bitcoin trading bot using data science, deep learning, and time-series analysis concepts.

ADDITIONAL PROFESSIONAL EXPERIENCE

Content Reviewer and Writer, GeeksforGeeks, Remote Jan 2023 – Dec 2023

- Reviewed technical articles and code examples for correctness and authored tutorials on computer science topics.

Software Developer, Seva LLP, Ajmer, India Feb 2023 – Aug 2023

- Designed, documented, and maintained Zoho Creator applications; resolved bugs and implemented new features in collaboration with cross-functional teams.

Data Science Intern, Elamigo Ecom Pvt. Ltd., Remote Dec 2022 – Feb 2023

- Developed data-driven solutions using Python, R, and SQL.

WordPress Developer, Unvoiced Media and Entertainment Pvt. Ltd., Remote Apr 2021 – Oct 2022

- Customized themes, maintained site performance, and ensured website functionality and security.

Subject Matter Expert, Chegg, India Nov 2020 – Jun 2021

- Answered 1000-plus computer science questions with detailed and accurate technical explanations.

IT Intern, Seva Enterprise Limited, Ajmer, India Jan 2021 – Mar 2021

- Developed Zoho-based solutions and contributed to web development tasks for the organization.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, SQL, JavaScript, HTML, CSS
Frameworks and Libraries: PyTorch, TensorFlow, scikit-learn, OpenCV, React
Research Areas: Machine Learning, Deep Learning, Computer Vision, Medical Imaging, Vision-Language Models, Natural Language Processing, Data Mining
Tools and Platforms: Git, MATLAB, LaTeX, WordPress, Zoho Creator
Data Science Skills: Data Analysis, Statistical Modeling, Hypothesis Testing, Time-Series Analysis, Feature Engineering, Data Visualization

HONORS AND AWARDS

- Qualified NET-JRF (December 2022).

- Qualified NET (June 2022).
- Scored 93.13 percentile in GATE 2023.
- Department Representative, Department of Computer Science, Central University of Rajasthan.
- Mega Mess Committee Member, Central University of Rajasthan.
- Member, Svatva Foundation (NGO).

CERTIFICATIONS

- Machine Learning Specialization, Coursera.
- Machine Learning Specialization, Internshala.
- Google IT Support Professional Certificate, Coursera.
- Programming, Data Structures and Algorithms Using Python, NPTEL.
- Programming in Java, NPTEL.

REFERENCES

Prof. Amit Sethi
Professor, Department of Electrical Engineering
Indian Institute of Technology Bombay
Email: asethi@iitb.ac.in Mobile: 8011017687

Dr. Krishna Kumar Mohbey
Assistant Professor, Department of Computer Science
Central University of Rajasthan
Email: kmohbey@curaj.ac.in Mobile: 9826968036

Dr. Nishtha Kesswani
Associate Professor, Department of Data Science and Analytics
Central University of Rajasthan
Email: nishtha@curaj.ac.in Mobile: 9414222654